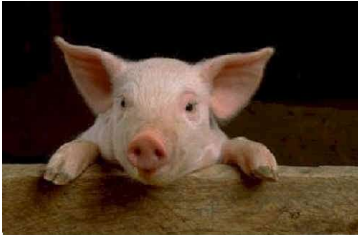


Programming Project: Pig

(Game Rules for Pig were taken from http://boardgames.about.com/od/dicegames/a/pig_rules.htm)

Overview



<http://www.all-creatures.org/anex/pig-01.html>



<http://www.grand-illusions.com>

Need to pass the time? Grab a die and play some Pig. The game play is as follows: a player will roll the 6-sided die. If a one is rolled, the player busts and the play moves to the next player, without adding any points to his/her score. Otherwise, the current roll value is added to a running total and the player may choose to roll again (to continue adding to the running total) or they may hold (keep the current running total and add it to their overall score). Note that a player only adds points to their score if they choose to hold on a

particular turn. That means, if a player has a running total of 15, but rolls a 1 next, that player loses the 15 points built up in that set of rolls and no additional points are added to his/her overall score. A player wins when they have earned 100 or more points. In a case where player 1 hits 100 first, then let player 2 have a turn to try and match/win. If they both are over 100, the higher score wins! This

game has many other variations some of which can be found at this website: <http://cs.gettysburg.edu/projects/pig/piglinks.html>

Objectives

- To practice using the Scanner class to retrieve console input
- To practice using loops appropriately in a particular context
- To practice writing classes and using helper methods

What to Do

You (and possibly a partner) are to write a program that will allow two human players to play Pig. When the game starts, a welcome message and instructions for the game will appear. The players will then be asked to enter their names. The computer will prompt each person to roll to see who goes first. On each turn, the current player will roll and be able to decide to roll again (1) or hold (2). If they hold, their running total will be added to their score and their turn ends. If they roll again, as long as they do not roll a 1, the number on the die will be added to their running total. If a 1 is rolled, a message telling the player they busted will display and their turn will end. In between each player's turn, the game should display an updated "scoreboard". The first player to bank 100 points before the other wins! [If both bank over 100 points in the same turn, then the one with the higher score wins!] At the end of the game, the players will be asked if they want to play again. If so, the game resets and begins again.

Turn in two files, **PlayPig.java** and **Player.java** when you have finished the assignment. Turn them in on the google classroom assignment and also turn in a printed copy of your code. Put your names in a comment at the top of the classes. Code that is identical to other group's code will be penalized.

AP Computer Science

Helpful Hints

- Start with the English – build pseudocode. Natural tasks can be turned into methods!
- **Write a Player class. Every Player will have a name and a score and will be able to take a turn.**
- A Pig game has a couple players that alternate taking turns until some win condition happens. Ultimately, we “play” a Pig game. Be sure to break down your tasks into appropriate helper methods as you go.

Requirements

- **You must put comments at the top of each method** in PlayPig.java to explain what each method does
- **Your game must be fully functional:** all the rules for busting, rolling again, holding and winning apply and work as specified
- **When a player has won, a message should print saying so**
- **You must use helper methods**, only the `play()` method will be called from main method
- For full credit, **you must have at least the following three methods** other than `main` in your program:
 - A method called `play()`
 - A method called `welcome()` that prints out the introduction information
 - A method called `displayScoreBoard` that takes two Player objects as parameters which prints out the scoreboard info after each round

Grading Rubric

Start of game displays welcome and instructions & asks for names of the players

Players will roll at start to see who goes first successfully

Bust, Roll Again, and Hold actions are processed properly

Program successfully determines (and outputs) a winner

Style of program is satisfactory – good indentation, naming, use of helper methods, comments, etc.

Finished early? Try these extensions:

- Ask user if they want to play again when a game is finished. If yes, allow users to play a new game.
- Allow the user will be able to select how many players will be participating (between 2 and 6 players). The game will then play according to their selection.
- Add a game AI:
 - Allow the player(s) to add computer players to the game
 - Allow for the computer to play according to various strategies

Two sample games are on the following pages.

Beginning of One Game

*** Welcome to the Game called Pig! ***

Each player will take turns rolling a dice and the first to reach 100 wins.
If a 1 is rolled on a turn, the player busts and earns nothing for their turn.
If a 1 is NOT rolled the player may choose to keep rolling the dice and
the total of the rolls of their turn will be added to their score.

Enter player 1's name: Tom
Enter player 2's name: Jerry

Tom ->type any letter to roll the dice:r
You rolled a 6

Jerry ->type any letter to roll the dice:r
You rolled a 6
You tied - Tom will go first

The rest of the game is not shown

A Complete Game

*** Welcome to the Game called Pig! ***

Each player will take turns rolling a dice and the first to reach 100 wins.
If a 1 is rolled on a turn, the player busts and earns nothing for their turn.
If a 1 is NOT rolled the player may choose to keep rolling the dice and
the total of the rolls of their turn will be added to their score.

Enter player 1's name: Tom
Enter player 2's name: Jerry

Tom ->type any letter to roll the dice:r
You rolled a 5

Jerry ->type any letter to roll the dice:r
You rolled a 6
Jerry rolled a bigger # and will go first

***Jerry ->type any letter to roll the dice:r
Jerry - you rolled a 4
Your current round total is now: 4
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 4
Your current round total is now: 8
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 4
Your current round total is now: 12
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 2
Your current round total is now: 14
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 3
Your current round total is now: 17
Jerry -> ROLL AGAIN? Type y or n:y

Jerry - you rolled a 2
Your current round total is now: 19
Jerry -> ROLL AGAIN? Type y or n:n
Jerry your turn is over - 19 points added - your score is now: 19

***Tom ->type any letter to roll the dice:r
Tom - you rolled a 4
Your current round total is now: 4
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 3
Your current round total is now: 7
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 6
Your current round total is now: 13
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 5
Your current round total is now: 18
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 5
Your current round total is now: 23
Tom -> ROLL AGAIN? Type y or n:n
Tom your turn is over - 23 points added - your score is now: 23
----- Scoreboard -----
Player 1: Tom score: 23
Player 2: Jerry score: 19

***Jerry ->type any letter to roll the dice:r
Jerry - you rolled a 5
Your current round total is now: 5
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 3
Your current round total is now: 8
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 1

Bust! Your turn is over. Nothing was added to your score.

***Tom ->type any letter to roll the dice:r
Tom - you rolled a 2
Your current round total is now: 2
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 1

Bust! Your turn is over. Nothing was added to your score.

----- Scoreboard -----
Player 1: Tom score: 23
Player 2: Jerry score: 19

***Jerry ->type any letter to roll the dice:r
Jerry - you rolled a 5
Your current round total is now: 5
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 1

Bust! Your turn is over. Nothing was added to your score.

***Tom ->type any letter to roll the dice:r
Tom - you rolled a 1

Bust! Your turn is over. Nothing was added to your score.

----- Scoreboard -----

Player 1: Tom score: 23

Player 2: Jerry score: 19

***Jerry ->type any letter to roll the dice:r

Jerry - you rolled a 1

Bust! Your turn is over. Nothing was added to your score.

***Tom ->type any letter to roll the dice:r

Tom - you rolled a 6

Your current round total is now: 6

Tom -> ROLL AGAIN? Type y or n:y

Tom - you rolled a 5

Your current round total is now: 11

Tom -> ROLL AGAIN? Type y or n:y

Tom - you rolled a 6

Your current round total is now: 17

Tom -> ROLL AGAIN? Type y or n:y

Tom - you rolled a 1

Bust! Your turn is over. Nothing was added to your score.

----- Scoreboard -----

Player 1: Tom score: 23

Player 2: Jerry score: 19

***Jerry ->type any letter to roll the dice:r

Jerry - you rolled a 4

Your current round total is now: 4

Jerry -> ROLL AGAIN? Type y or n:y

Jerry - you rolled a 6

Your current round total is now: 10

Jerry -> ROLL AGAIN? Type y or n:y

Jerry - you rolled a 1

Bust! Your turn is over. Nothing was added to your score.

***Tom ->type any letter to roll the dice:r

Tom - you rolled a 6

Your current round total is now: 6

Tom -> ROLL AGAIN? Type y or n:y

Tom - you rolled a 5

Your current round total is now: 11

Tom -> ROLL AGAIN? Type y or n:n

Tom your turn is over - 11 points added - your score is now: 34

----- Scoreboard -----

Player 1: Tom score: 34

Player 2: Jerry score: 19

***Jerry ->type any letter to roll the dice:r

Jerry - you rolled a 2

Your current round total is now: 2

Jerry -> ROLL AGAIN? Type y or n:y

Jerry - you rolled a 5

Your current round total is now: 7

Jerry -> ROLL AGAIN? Type y or n:y

Jerry - you rolled a 5

Your current round total is now: 12

Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 1

Bust! Your turn is over. Nothing was added to your score.

***Tom ->type any letter to roll the dice:r
Tom - you rolled a 5
Your current round total is now: 5
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 3
Your current round total is now: 8
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 3
Your current round total is now: 11
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 2
Your current round total is now: 13
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 3
Your current round total is now: 16
Tom -> ROLL AGAIN? Type y or n:n
Tom your turn is over - 16 points added - your score is now: 50
----- Scoreboard -----
Player 1: Tom score: 50
Player 2: Jerry score: 19

***Jerry ->type any letter to roll the dice:r
Jerry - you rolled a 6
Your current round total is now: 6
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 6
Your current round total is now: 12
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 6
Your current round total is now: 18
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 4
Your current round total is now: 22
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 3
Your current round total is now: 25
Jerry -> ROLL AGAIN? Type y or n:n
Jerry your turn is over - 25 points added - your score is now: 44

***Tom ->type any letter to roll the dice:r
Tom - you rolled a 5
Your current round total is now: 5
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 3
Your current round total is now: 8
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 3
Your current round total is now: 11
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 6
Your current round total is now: 17
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 6
Your current round total is now: 23

Tom -> ROLL AGAIN? Type y or n:n
Tom your turn is over - 23 points added - your score is now: 73
----- Scoreboard -----
Player 1: Tom score: 73
Player 2: Jerry score: 44

***Jerry ->type any letter to roll the dice:r
Jerry - you rolled a 5
Your current round total is now: 5
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 6
Your current round total is now: 11
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 2
Your current round total is now: 13
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 3
Your current round total is now: 16
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 1

Bust! Your turn is over. Nothing was added to your score.

***Tom ->type any letter to roll the dice:r
Tom - you rolled a 1

Bust! Your turn is over. Nothing was added to your score.

----- Scoreboard -----
Player 1: Tom score: 73
Player 2: Jerry score: 44

***Jerry ->type any letter to roll the dice:y
Jerry - you rolled a 3
Your current round total is now: 3
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 6
Your current round total is now: 9
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 5
Your current round total is now: 14
Jerry -> ROLL AGAIN? Type y or n:y
Jerry - you rolled a 1

Bust! Your turn is over. Nothing was added to your score.

***Tom ->type any letter to roll the dice:r
Tom - you rolled a 2
Your current round total is now: 2
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 4
Your current round total is now: 6
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 4
Your current round total is now: 10
Tom -> ROLL AGAIN? Type y or n:y
Tom - you rolled a 2
Your current round total is now: 12
Tom -> ROLL AGAIN? Type y or n:n
Tom your turn is over - 12 points added - your score is now: 85

----- Scoreboard -----

Player 1: Tom score: 85

Player 2: Jerry score: 44

***Jerry ->type any letter to roll the dice:r

Jerry - you rolled a 2

Your current round total is now: 2

Jerry -> ROLL AGAIN? Type y or n:y

Jerry - you rolled a 5

Your current round total is now: 7

Jerry -> ROLL AGAIN? Type y or n:y

Jerry - you rolled a 6

Your current round total is now: 13

Jerry -> ROLL AGAIN? Type y or n:y

Jerry - you rolled a 4

Your current round total is now: 17

Jerry -> ROLL AGAIN? Type y or n:n

Jerry your turn is over - 17 points added - your score is now: 61

***Tom ->type any letter to roll the dice:r

Tom - you rolled a 5

Your current round total is now: 5

Tom -> ROLL AGAIN? Type y or n:y

Tom - you rolled a 4

Your current round total is now: 9

Tom -> ROLL AGAIN? Type y or n:y

Tom - you rolled a 3

Your current round total is now: 12

Tom -> ROLL AGAIN? Type y or n:y

Tom - you rolled a 5

Your current round total is now: 17

Tom -> ROLL AGAIN? Type y or n:y

Tom - you rolled a 5

Your current round total is now: 22

Tom -> ROLL AGAIN? Type y or n:n

Tom your turn is over - 22 points added - your score is now: 107

Tom Wins!!

----- Scoreboard -----

Player 1: Tom score: 107

Player 2: Jerry score: 61